

Aura Artifacts

E-Commerce Platform for Handcrafted Gemstone Jewelry

Project Report

2026

1. Overview

Aura Artifacts is an e-commerce platform for handcrafted, customizable gemstone jewelry. The platform lets customers browse a product catalog, design custom bracelets by combining gemstones and charms, manage a shopping cart, complete checkout, and track their orders. An administrative panel supports catalog management, order processing, customer management, and store analytics.

The repository contains two parts: a static HTML and CSS prototype that establishes the page structure and visual design, and the complete implementation built with PHP and MySQL.

2. Features

Storefront

- Home page, collections page with category filtering, and product detail pages
- Live product search
- Gemstone meanings guide and a gemstone-matching quiz
- Bracelet size guide
- Accessibility information page

Custom bracelet builder

- Selection of gemstones and charms across thirteen charm categories
- Real-time preview and price calculation as pieces are added
- Saved designs can be added to the cart like a regular product

Customer accounts

- Registration and login with a seven-day remember-me option
- Profile management and order history
- Wishlist

Shopping and checkout

- Shopping cart with quantity updates
- Checkout with shipping and payment details
- Order confirmation
- Newsletter subscription and a contact form with an embedded map

Admin panel

- Dashboard with sales analytics and store metrics
- Product, stock, order, customer, message, and subscriber management
- Site settings

Other

- Dark mode
- Live currency conversion

3. Technology

Layer	Technology
Frontend	HTML5, CSS3, vanilla JavaScript
Backend	PHP, using mysqli with prepared statements
Database	MySQL / MariaDB
Authentication	Session-based, bcrypt password hashing, remember-me tokens
External services	ExchangeRate-API for currency conversion, Google Maps Embed for the contact page

4. Database

The database consists of sixteen tables covering catalog data, customer accounts, orders, and site content.

Table	Purpose
customer	Customer accounts
admin	Administrator accounts
category	Product categories
product	Catalog products
gemstone	Gemstones available for the bracelet builder
charm	Charms available for the bracelet builder
braceletpreview	Saved custom bracelet designs

<code>cart , cart_item</code>	Shopping cart and its line items
<code>order , order_item</code>	Placed orders and their line items
<code>wishlist</code>	Customer wishlists
<code>contact_message</code>	Contact form submissions
<code>newsletter</code>	Newsletter subscribers
<code>password_reset_request</code>	Password reset requests
<code>notification_read</code>	Read state of admin notifications

The full entity-relationship diagram is included in the repository documentation.

5. Repository structure

Path	Contents
<code>01-frontend-html-css-erd/</code>	Static HTML and CSS prototype, plus the database ERD and its explanation
<code>02-full-project/source/</code>	The full PHP and MySQL application
<code>02-full-project/documentation/</code>	Project report, JavaScript and API documentation, and the full-system ERD

6. Running the project

The full implementation requires PHP 8 with the `mysqli` extension, and MySQL or MariaDB.

- Create a database and import `02-full-project/source/database.sql`
- Copy `02-full-project/source/.env.example` to `.env` and set the database credentials
- Serve `02-full-project/source/`, for example with `php -S localhost:8000`

The static prototype in `01-frontend-html-css-erd/` requires no setup and can be opened directly in a browser.